

FILIPPO MARCANTONI

Robotics Software Engineer | Computer Vision & 3D Perception | Robot Learning | Physical AI
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SUMMARY

Robotics engineer with an M.S. in Robotics Engineering from Worcester Polytechnic Institute and hands-on R&D experience across robotics software, computer vision, robot learning, industrial automation, teleoperation, swarm robotics, and generative AI. Experienced in building and integrating systems across perception, simulation, learning, testing, and real-world deployment. Seeking robotics software, robot learning, computer vision, or physical AI roles with the goal of contributing to practical robotic systems and continuing to grow as a full-stack robotics engineer.

EDUCATION

Worcester Polytechnic Institute (WPI)

M.S. Robotics Engineering | GPA 3.90/4.00

B.S. Robotics Engineering | GPA 3.89/4.00, Summa Cum Laude

Worcester, MA, USA

Aug 2024 – May 2026

Aug 2021 – May 2025

EXPERIENCE

AI & Industrial Robotics R&D Intern | B&R Industrial Automation, Austria

Jun 2025 – Aug 2025

- Developed an LLM-based code-generation copilot for B&R Automation Studio using Transformers, RAG, Azure AI retrieval, and Motion ST templates to support industrial robot programming and automation workflows.
- Designed a robotics-specific retrieval backend with domain-aware metadata extraction and syntax-aware chunking for Structured Text and Motion ST code, improving the quality of retrieved context across large automation codebases.

Machine Vision & AI R&D Intern | Makro Labelling SRL, Italy

May 2024 – Aug 2024

- Developed C++ and Python machine-vision pipelines for bottle orientation, label alignment, and defect inspection on high-speed labeling lines.
- Benchmarked classical and deep-learning vision models across bottle formats, lighting conditions, and label shifts to identify failure modes, validate robustness, and improve automated quality-control performance.

RESEARCH

VR-Based Multimodal Teleoperation Interface | HiRo Lab, WPI

Jan 2025 – May 2025

- Developed a human-in-the-loop VR teleoperation system for a robotic nursing assistant using Unity, ROS, Meta Quest head-motion mirroring, real-time multi-camera streaming, Whisper voice commands, and ZED-Mini 3D object detection.
- Integrated viewpoint control, speech-based camera switching, and perception-driven camera selection to improve operator situational awareness during remote robotic manipulation.

PiELo: Reactive Swarm Programming Language | NEST Lab, WPI

Aug 2024 – May 2025

- Designed PiELo, a C++ reactive swarm-programming language with consensus and virtual stigmergy as first-class primitives for distributed robotic coordination.
- Validated distributed swarm behaviors in ARGoS3 simulation and on Khepera IV robots with Vicon motion tracking.

STL-to-Fluoroscopy 2D-3D Registration | FuTURE Lab, WPI

Aug 2025 – Dec 2025

- Designed a self-supervised fluoroscopy-to-volume registration workflow using DiffDRR, 3D Slicer, and DICOM processing for image-guided endovascular procedures.
- Validated patient-specific pose estimation on a 3D-printed aorta phantom without ground-truth labels.

SELECTED PROJECTS

Robot Learning: Reinforcement & Imitation Learning

- Implemented policy iteration, REINFORCE, A2C, and A3C for simulated control and robotic manipulation tasks, including LunarLander and PyBullet Kuka vision-based grasping, reaching 55% grasp success with a CNN actor-critic policy.
- Trained imitation-learning policies in robomimic on 60 PickPlaceCan demonstrations, improving success from 0.02 with baseline behavioral cloning to 0.80 with LSTM behavioral cloning and 0.90 with diffusion policy.

Classical MSCKF & Deep Visual-Inertial Odometry

- Implemented a stereo MSCKF visual-inertial odometry backend with IMU propagation, camera-state cloning, stereo feature tracking, and nullspace EKF updates, evaluated on EuRoC and RPG datasets.
- Trained IMU-only, vision-only, and fused deep VIO networks on synthetic Blender trajectories, evaluating sensor-fusion behavior with geodesic SO(3) and trajectory-composition losses.

Monocular 3D Driving-Scene Reconstruction

- Built a monocular 3D perception pipeline integrating object detection, segmentation, depth estimation, pose estimation, and optical flow for autonomous-driving scene understanding.
- Reconstructed 3D driving scenes in Blender with lanes, vehicles, pedestrians, traffic objects, orientation, motion state, and collision-risk estimation.

3D Scene Reconstruction: Classical SfM & NeRF

- Implemented Structure-from-Motion from scratch, including SIFT tracking, RANSAC, fundamental matrix estimation, essential-matrix recovery, cheirality checks, PnP-RANSAC, triangulation, and bundle adjustment.
- Built a PyTorch NeRF with positional encoding, 5D MLPs, hierarchical sampling, and differentiable volume rendering.

TECHNICAL SKILLS

Languages: Python, C++, C, C#, MATLAB, Java, JavaScript, Buzz.

Robotics & Simulation: ROS/ROS2, Gazebo, MuJoCo, PyBullet, ARGoS3, robosuite, robomimic, Unity, Blender, Vicon, VR/XR, Dynamixel, ZED SDK, Automation Studio, 3D Slicer.

Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, TensorRT, CUDA, Transformers, LLMs, RAG, Reinforcement Learning, Imitation Learning, Behavioral Cloning, Diffusion Policy, Actor-Critic Methods.

Computer Vision & 3D Perception: OpenCV, SLAM, Visual-Inertial Odometry, MSCKF, SfM, NeRF, RANSAC, PnP, Bundle Adjustment, Homography, Object Detection, Segmentation, Monocular Depth, Pose Estimation, Optical Flow, Point Clouds.

Tools & Infrastructure: Linux, Git, Docker, Jupyter, Azure AI, AWS, Simulink, React, Node.js, PostgreSQL, MySQL, HPC/Slurm.

CAD & Design: SolidWorks, Fusion 360.